

JACOPO MANENTI

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EXPERIENCE

Financial Analyst – Market Operations, Innovation & Data Analytics

March 2025 – Present

European Central Bank

Frankfurt am Main, Germany

- Developed an AWS cloud-based RAG application for collateral eligibility serving ECB and NCB experts, and conducted ML-driven financial research, e.g. delivering econometric and ML-based recommendations for US bonds during monthly portfolio rebalancing.
- Built data pipelines using dbt for market operations, processing daily repo and money market transactions, integrating data from internal databases, Bloomberg, OneTick, and other external data providers.
- Active presenter and participant at the Market Artificial Intelligence Network (MAIN), delivering presentations showcasing innovative AI projects to national central banking professionals across the Eurosystem.
- Serve as peer-to-peer trainer for ECB staff, conducting 5 training sessions on AI (LLMs, RAGs) for 100+ employees.

Junior Analyst – Master Servicer for NPL Loans Portfolio

September 2022 – September 2023

Zenith Global S.p.A.

Milan, Italy

- Conducted periodic analyses of non-performing loan (NPL) portfolios, managing over €1 billion in Gross Book Value (GBV).
- Managed investor reporting processes on a monthly, quarterly, and semiannual basis, utilizing Qlik for data visualization to deliver accurate financial updates on portfolio status and risk exposure to stakeholders.
- Ensured regulatory compliance by adhering to the requirements of the prospectus and credit transfer agreements, including detailed reporting for the Bank of Italy.

EDUCATION

Università di Trento & ELTE Budapest

Trento & Budapest

Double M.Sc. in Computer Science (EIT Digital), Fintech – 110 cum laude (IT), 5/5 (HU)

September 2023 – July 2025

Università Cattolica del Sacro Cuore

Milan, Italy

B.Sc. in Economics, Organizations and Markets – 110 cum laude

September 2019 – July 2022

AWARDS AND EXTRACURRICULAR ACTIVITIES

SurrealDB x LangChain Hackathon, London

April 2026

- Developed an AI-powered platform with a chat-like interface to help companies perform supply chain due diligence on suppliers and partners, experimenting with agents and knowledge graphs over a 3-day hackathon.

1st Place – ECB Hackathon “From News to AI Forecasts”

November 2025

- Secured first place developing novel sentiment-driven economic indicators using the GDELT Global Knowledge Graph, integrating them with traditional indicators to refine Chronos II, an AI time-series foundation model, to improve medium-term forecasts of negotiated wages in the Eurozone.

Summer School in Fintech, Universidad Politécnica de Madrid

July 2024

- Completed a two-week summer school program focused on Fintech, Machine Learning applied to finance, Fintech regulation, and blockchain.

Startup Lab (CLab Trento)

March 2024 – June 2024

- Developed a 3D fitting room plugin for apparel marketplaces and created the business model for the startup. Pitched the idea during the Innovation Challenge at CLab, securing 4th place and earning a two-week program at the HIT accelerator in Trento.

EIT Student Ambassador

November 2023 – Present

- Promoting the EIT Digital program and participating in Tech events to engage students and potential business sponsors.

PROJECTS

RAG System for Collateral Eligibility (ECB)

2025 – 2026

- Developed an LLM-powered RAG chatbot on ECB collateral eligibility and haircut rules, integrating public and internal documentation into a curated knowledge base with a retrieval pipeline grounding responses in official sources.
- Deployed in an AWS environment and used daily by ECB and NCB collateral experts.

Settlement Monitoring Dashboard (ECB)

2025

- Built a Tableau dashboard for the back office tracking each repo transactions (settled amount, settlement status, clearing ID), reconciling treasury management system data against reports from clearing houses.

ZK Proof & Decision Tree-Based Algorithmic Trading on S&P 500 | [Link](#)

January 2025

- Engineered a high-frequency trading bot for S&P 500 stocks and ETFs (Alpaca API) with automated bracket orders, real-time Grafana/InfluxDB monitoring, backtesting framework, and Docker containerization.
- Pioneered integration of Zero-Knowledge Proofs in algorithmic trading via Zokrates smart contracts, enabling trade verification while preserving algorithmic privacy.

Graph Neural Network-Based Fraud Detection in Cryptocurrency Transactions | [Link](#)

December 2024

- Engineered GCN achieving 95% AUC-ROC for Bitcoin fraud detection on the Elliptic Data Set over different seeds.
- Outperformed traditional GCN baseline by +12%, demonstrating 97% accuracy and 87% precision in illicit transaction identification, implementing advanced architectures like GAT, GraphSMOTE and GNN Transformer.

NFT Marketplace for Music Royalties Distribution | [Link](#)

August 2024

- Designed a blockchain platform for small record companies to monetize music rights through NFTs, implementing 3 smart contracts in Solidity for NFT creation, marketplace functionality, and royalty management.
- Integrated Web3 technologies, including MetaMask and IPFS via Pinata, and built the frontend using Next.js and TypeScript.

Tail Distribution Analysis on FTSE MIB | [Link](#)

February 2024

- Performed a tail distribution analysis on FTSE MIB’s multi-day returns (up to 40 days) fitting the power law and exponentially decaying distribution, based on Roos, T. (2021), “Empirical Study of Tail Distributions of Multi-Day Index Returns.”

SKILLS

- Programming Languages:** Python, SQL, R, Java, MATLAB, Solidity
- Data Science & Tools:** dbt, Kafka, Tableau, Power BI, Qlik, LGTM Stack, Oracle DB, AWS Cloud, Excel
- Machine Learning & AI:** PyTorch, LLMs, RAG, LangChain, LangGraph, AWS Bedrock
- Finance & Blockchain:** Bloomberg Terminal & API, OneTick
- Languages:** Italian (Native), English (Proficient, IELTS C1), Spanish (Intermediate)